Synopsis

ON

KRISHI SETU

Research Proposal Submitted in Partial Fulfilment of the Requirements for the BTech Degree

In

Department Of Computer Science and Business Systems

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[2025-2026]

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Abstract

Agricultural markets in developing economies continue to face structural inefficiencies due to fragmented supply chains, a complete lack of price transparency, and limited direct interaction between farmers and end buyers. Farmers often rely on conventional, localized methods for crop sales and selection due to insufficient access to real-time market demand information, resulting in reduced profitability and a heavy reliance on intermediaries.

This research proposes **Krishi Setu**, a novel, location-aware digital marketplace that enables direct, real-time communication between farmers and buyers. The core intellectual property of this project lies in its unique conceptual framework: integrating an offline-first architecture (via local storage) with real-time negotiation tools (socket-based chat) specifically tailored for rural connectivity challenges. Built on modern technologies like React Native and Node.js, the system empowers farmers to list crops while buyers can search based on geolocation and specific requirements. The proposed solution is designed to be highly scalable, lightweight, and accessible, contributing towards building a transparent, efficient, and technology-driven agricultural ecosystem.

1 Introduction

Agriculture plays a crucial role in economic development; however, the sector is deeply affected by systemic inefficiencies that bottleneck productivity and profitability. One of the most significant challenges is the lack of a structured, accessible marketplace that facilitates direct interaction between producers and consumers. This vacuum leads to an absolute dependency on intermediaries and commission agents who artificially influence pricing, ultimately reducing the farmer's net income.

Another critical issue is the absence of reliable, data-driven market information. Farmers often make critical harvesting and pricing decisions based on traditional practices rather than real-time demand, resulting in mismatched supply and demand. While digital platforms have the potential to transform these practices, existing solutions primarily focus on B2B logistics or one-way advisory services, completely failing to provide integrated, real-time marketplace functionality for the individual farmer.

Krishi Setu is proposed as a unified conceptual system that combines direct marketplace access, location-based logistics, and real-time communication. The system's primary innovation is its idea-driven approach to rural e-commerce, empowering farmers with direct negotiation capabilities and improved market efficiency without the need for high-end digital infrastructure.

1.1 Background of the Study

Agricultural markets have traditionally operated through informal and unstructured physical systems (Mandis) where farmers depend heavily on local agents. These intermediaries often dictate pricing based on opaque criteria, leaving farmers with virtually no control over the true market value of their produce.

In recent years, while digital platforms have emerged, their adoption in rural areas remains severely limited. This is primarily due to usability challenges, high data consumption, and a lack of offline capabilities. Most current platforms fail to provide real-time, profile-to-profile interaction. The concept of Krishi Setu was developed specifically to address these challenges by creating a direct, transparent, and digitally lightweight marketplace that connects farmers and buyers efficiently, even in areas with fluctuating internet connectivity.

2 Research Objectives and Approach

2.1 Research Objective

The primary objective of this research is to conceptualize and develop a digital platform that completely bypasses traditional intermediaries, enabling direct farmer-buyer interaction to support intelligent, fair-trade agricultural commerce.

The specific objectives include:

- **Eliminating Dependency on Intermediaries:** Creating a digital negotiation space to facilitate direct peer-to-peer transactions.
- **Improving Price Transparency:** Implementing a real-time chat module for open negotiation and fair pricing.
- **Enhancing Rural Accessibility:** Developing an offline-first mobile architecture (using technologies like MMKV) to ensure usability in low-network zones.
- **Logistical Optimization:** Utilizing geolocation services to match local supply with nearby demand, reducing transportation overheads.
- **Developing a Scalable Platform:** Engineering a robust backend using Node.js and MongoDB to support rapid scaling and data management.
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2.2 Research Methodology

The research follows a structured, agile methodology encompassing requirement analysis, system architecture design, core implementation, and rigorous evaluation.

Initially, the core agricultural problems and user requirements were analysed to conceptualize the specific features needed for rural adoption. The system architecture was then designed, separating

the platform into distinct modules: User Authentication, Farmer/Buyer Dashboards, Product Listing, and a Real-Time Communication Engine (Socket.io).

The application is being developed using React Native for cross-platform mobile efficiency, integrated with a cloud-based Node.js backend. A significant focus of the methodology is placed on performance optimization, utilizing tools like Flash List for smooth rendering on low-end devices. The system will be evaluated under simulated rural network conditions to test its offline-first capabilities and real-time synchronization.

3 Review of Work already done on the subject

Extensive market and literature research was conducted on existing aggrotech platforms, including Agri Bazaar, DeHaat, and Ninjacart. While these platforms have made strides in modernizing agriculture, the analysis reveals significant conceptual gaps:

- **Focus on Supply Chain, Not Farmers:** Existing systems primarily optimize B2B supply chain logistics rather than empowering individual farmers in a C2C or B2C model.
- **Lack of Direct Negotiation:** Current applications act as digital middlemen themselves, offering fixed prices rather than providing a real-time communication channel for farmers and buyers to negotiate directly.
- **Poor Rural Optimization:** Existing platforms are often heavy and require persistent high-speed internet, lacking the offline-first architecture necessary for deep rural penetration.

Krishi Setu addresses these exact gaps by introducing a novel framework that prioritizes direct communication and lightweight, offline-capable performance over generalized logistics.

4 Current Work and Preliminary Results

4.1 Problem Definition

Farmers lack access to an optimized, mobile-first digital platform that enables direct communication with end-buyers and provides real-time market insights. Furthermore, the lack of applications tailored for poor network conditions exacerbates the digital divide, forcing continued dependency on exploitative intermediaries.

4.2 Problem Statement

There is a critical need to conceptualize and develop an offline-capable, highly optimized digital system that establishes direct, real-time interaction between farmers and buyers, thereby democratizing agricultural trade and supporting data-driven decision-making.

4.3 System Workflow Description

The system operates on a decentralized, user-centric workflow. It begins with role-based registration and authentication (Farmer or Buyer). Farmers upload their crop details, including

images, expected pricing, and location, which are processed and stored in a highly scalable NoSQL database (MongoDB).

Buyers utilize the platform to search for specific crops using dynamic filters such as geolocation (maps integration) and price range. Once a suitable listing is identified, the buyer initiates direct communication via the integrated Socket.io chat module. This allows for real-time negotiation, enabling transactions to be finalized fairly and efficiently. Crucially, the system utilizes local storage (MMKV) to cache data, ensuring that the app remains functional and navigable even when the user temporarily loses internet connectivity, syncing automatically once the connection is restored.

5 Work Plan and Implications

The project is divided into agile developmental phases: Ideation & Research, System Design & Architecture, Core Development (Frontend/Backend integration), Advanced Features (Location & Real-time Chat), and UI/UX Optimization. Each phase ensures systematic development and rigorous debugging (e.g., handling ADB connectivity and multi-device rendering).

The implications of this system are substantial. By protecting and executing this novel conceptual framework, Krishi Setu is expected to drastically improve market transparency, maximize farmer profit margins, and establish a new baseline for how agricultural e-commerce should operate in developing regions.

5.1 Study Hypotheses

- **H_0 (Null Hypothesis):** The implementation of an offline-first digital marketplace with integrated real-time communication does not significantly improve market efficiency, price transparency, or farmer income.
- **H_1 (Alternate Hypothesis):** The proposed system (Krishi Setu) significantly improves market efficiency and farmer income by eliminating intermediaries through direct digital interaction and optimized localized discovery.

5.2 Detailed Explanation of Null Hypotheses

The null hypothesis assumes that introducing this technological framework will yield no measurable impact compared to traditional physical markets. This hypothesis will be rigorously evaluated by analysing system performance data, transaction negotiation times, user adoption rates, and qualitative feedback from rural farmers regarding ease of use and profit margin changes.

6 Need of the Proposed Research Work

The agricultural sector is at a critical juncture where conventional advisory platforms are no longer sufficient. There is an immediate need for integrated solutions that directly address the core issue

of market access and fair pricing. Existing systems are structurally limited, heavy, and often act as modern intermediaries.

Krishi Setu is necessary because it introduces a unified, ideologically novel platform that combines direct marketplace features with real-time communication and rural-friendly performance optimization. This research provides a scalable, sustainable blueprint that directly empowers the primary producers of the economy, ensuring they receive the true financial value of their labour.

7 References

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Signatures

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